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TREND-BASED MOCK PAPER

GATE Biotechnology

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Biotechnology
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Choose similar to 'catalyst':

(Numerical Answer Type — Enter the numerical value)

Section B: Biotechnology

Q2. [1-MCQ] [1 mark]

Normal distribution mean = median = ?

- (A) Mode
 - (B) Median
 - (C) Mean
 - (D) SD
-

Q3. [1-NAT] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
 - (B) 2
 - (C) 4
 - (D) 1
-

Q5. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908

- (B) 0.092
- (C) 0.762
- (D) 0.195

Q6. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q7. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q9. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)(1+z)}$

Q10. [1-NAT] [1 mark]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q12. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a, b]$ and differentiable on (a, b)
- (D) Both continuous and differentiable everywhere

Q13. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6

- (C) 36
- (D) 12

Q14. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q16. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q17. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q19. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q20. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q21. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0, 1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95

(D) 0.16

Q22. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q24. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q25. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q26. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q27. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C * x$
- (B) $y = C/x$
- (C) $y = C * x^2$
- (D) $x^2 + y^2 = C$

Q28. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q29. [2-NAT] [2 marks]

The vector (1,2,3) projected onto (2,1,0) is:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q31. [1-MCQ] [1 mark]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
 - (B) Nucleus
 - (C) Mitochondria only
 - (D) G1 phase
-

Q32. [1-MCQ] [1 mark]

Central dogma: reverse transcription is:

- (A) RNA->DNA
 - (B) DNA->RNA
 - (C) Protein->DNA
 - (D) RNA->Protein
-

Q33. [1-NAT] [1 mark]

Photosynthesis light reactions produce:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-MCQ] [1 mark]

Competitive enzyme inhibition: K_m

- (A) Increases
 - (B) Decreases
 - (C) Unchanged
 - (D) Zero
-

Q35. [1-MCQ] [1 mark]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
 - (B) 9:3:3:1
 - (C) 1:2:1
 - (D) 1:1:1:1
-

Q36. [2-NAT] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MCQ] [2 marks]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
- (B) Nucleus
- (C) Mitochondria only
- (D) G1 phase

Q38. [2-MSQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q39. [2-MCQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q40. [2-MCQ] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
- (B) 9:3:3:1
- (C) 1:2:1
- (D) 1:1:1:1

Q41. [2-MSQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q42. [2-MCQ] [2 marks]

Photosynthesis light reactions produce:

- (A) Glucose only
- (B) ATP and NADPH
- (C) CO₂
- (D) O₂ only

Q43. [1-MCQ] [1 mark]

Strongest acid:

- (A) HCl
- (B) HNO₃
- (C) H₂SO₄
- (D) HClO₄

Q44. [1-MCQ] [1 mark]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q45. [2-MCQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q46. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q47. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q48. [2-NAT] [2 marks]

Strongest acid:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-MSQ] [2 marks]

First-order half-life with $k = 0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q50. [2-MSQ] [2 marks]

Strongest acid:

- (A) HCl
- (B) HNO₃
- (C) H₂SO₄
- (D) HClO₄

Q51. [1-MCQ] [1 mark]

Law of segregation describes:

- (A) Alleles
- (B) Genes
- (C) Chromosomes
- (D) DNA

Q52. [1-NAT] [1 mark]

DNA stands for:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-MCQ] [1 mark]

ABO blood group has how many alleles?

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Q54. [1-MCQ] [1 mark]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q55. [1-MCQ] [1 mark]

Linked genes show:

- (A) Independent assortment
- (B) No recombination
- (C) Less than 50% recombination
- (D) Always 50%

Q56. [1-MCQ] [1 mark]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q57. [1-MCQ] [1 mark]

DNA B-form base pairs per turn:

- (A) 10
- (B) 10.4
- (C) 12
- (D) 8

Q58. [2-NAT] [2 marks]

Linked genes show:

(Numerical Answer Type — Enter the numerical value)

Q59. [2-MSQ] [2 marks]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q60. [1-MCQ] [1 mark]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond

(D) None

Q61. [1-NAT] [1 mark]

Binary fission in:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-NAT] [1 mark]

Batch process is:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Antibody produced by:

- (A) T cells
- (B) B cells
- (C) Macrophages
- (D) None

Q64. [1-NAT] [1 mark]

BLAST is for:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Chromosome number in daughter:

- (A) Halved
 - (B) Same
 - (C) Doubled
 - (D) Variable
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	C	1
3	1MCQ	A	1
4	1NAT	A	1
5	1MCQ	A	1
6	1NAT	A	1
7	1NAT	B	1
8	1MCQ	B	1
9	1NAT	A	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	A	1
13	1NAT	B	1

14	1MCQ	B	1
15	1NAT	A	1
16	1MCQ	A	1
17	1MCQ	A	1
18	1MCQ	B	1
19	1MCQ	C	1
20	1MCQ	D	1
21	1MCQ	C	1
22	1MCQ	A	1
23	1MCQ	A	1
24	1MCQ	B	1
25	1MCQ	B	1
26	1NAT	A	1
27	1MCQ	A	1
28	1MCQ	A	1

29	1MCQ	B	1
30	1NAT	A	1
31	2MCQ	C	2
32	2MCQ	C	2
33	2MCQ	D	2
34	2MCQ	B	2
35	2MCQ	A	2
36	2MCQ	B	2
37	2NAT	A	2
38	2MCQ	B	2
39	2MCQ	B	2
40	2NAT	C	2
41	2MCQ	A	2
42	2MCQ	B	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2NAT	C	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2MCQ	A	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2NAT	C	2
52	2NAT	B	2
53	2MCQ	B	2
54	2MSQ	A	2
55	2MCQ	A	2
56	2NAT	B	2
57	2NAT	D	2
58	2MCQ	A	2

59	2MCQ	B	2
60	2MSQ	A	2
61	2MSQ	A	2
62	2MSQ	A	2
63	2MSQ	D	2
64	2NAT	C	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Mendel | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Molecular Biology | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Biochemistry | Topic: Proteins | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Microbiology | Topic: Bacteria | Difficulty: difficult

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Bioprocess Engineering | Topic: Fermentation | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Immunology | Topic: Antibodies | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Bioinformatics | Topic: Sequence | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Cell Biology | Topic: Mitosis | Difficulty: easy

Q11.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q12.

Answer: A

Solution:

Reverse transcription: RNA→DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: difficult

Q13.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: easy

Q14.

Answer: B

Solution:

ABO: I^A, I^B, i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q15.

Answer: A

Solution:

$L\{e^{-(s+a)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q16.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q17.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q18.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: easy

Q19.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q20.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q21.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q22.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: A

Solution:

$$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q24.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q25.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q26.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x \cdot v$ and $y(1) = 0$ gives $y = 0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: A

Solution:

$f(x) = x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: A

Solution:

$$\text{PGF of Poisson}(\lambda): G(z) = E[z^X] = \exp(\lambda(z-1)).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q29.

Answer: B

Solution:

B-DNA: ~ 10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q30.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b)$. Then $f'(c) = 0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q34.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q35.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q37.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: B

Solution:

$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q45.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

$L\{t * f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q52.

Answer: B

Solution:

$AaBb \times AaBb$: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q53.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: easy

Q54.

Answer: A

Solution:

Reverse transcription: RNA $\rightarrow$ DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q55.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q57.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q58.

Answer: A

Solution:

Reverse transcription: RNA->DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q59.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q60.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q61.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q62.

Answer: A

Solution:

Reverse transcription: RNA->DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: easy

Q63.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q64.

Answer: C

Solution:

Rolle's: f continuous on [a,b], differentiable on (a,b), f(a)=f(b). Then f'(c)=0 for some c in (a,b).

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q65.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 29 questions, 49 marks
Biology: 12 questions, 19 marks
Chemistry: 8 questions, 14 marks
Genetics: 9 questions, 11 marks
General Aptitude: 1 questions, 1 marks
Biochemistry: 1 questions, 1 marks
Microbiology: 1 questions, 1 marks
Bioprocess Engineering: 1 questions, 1 marks
Immunology: 1 questions, 1 marks
Bioinformatics: 1 questions, 1 marks
Cell Biology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 19 | Moderate: 38 | Difficult: 8

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.